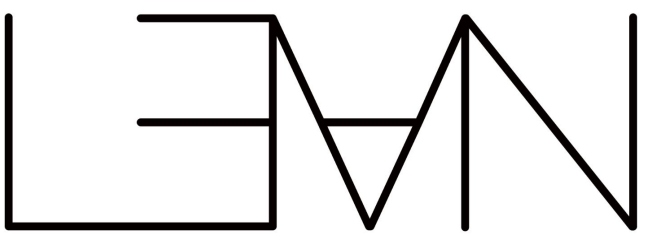


#33-R: Graph-Based Premise Recommendation in Lean/mathlib

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Introduction & Background

What is Lean 4



- Lean 4 is programming language used to formalize mathematics.
- Definitions, and theorems are written as **declarations**
- A proof is checked by Lean's kernel **automatically**

Problem:

Premise Recommendation

- Lean is a suitable setting for premise recommendation

[Kind] [name] [type] := value

Theorem_name

Proposition

Proof

Premise
(other theorem, definition...)

Data Information

Dataset & Graph Construction

- Detailed explanations
 - nodes: declarations
 - edges: $A \rightarrow B$ means A depends on B
 - Full graph vs Explicit-edge graph
- Mathlib
 - mathlib is Lean's large formal mathematics library

	kind	count	ratio
0	theorem	243725	0.7910
1	definition	48664	0.1579
2	abbrev	6667	0.0216
3	constructor	4755	0.0154
4	inductive	3813	0.0124
5	opaque	499	0.0016
6	quotient	3	0.0000
7	axiom	3	0.0000

```
#check Nat.add_comm
-- Nat.add_comm : ∀ (n m : ℕ), n + m = m + n
theorem add_comm_example (a b : ℕ) : a + b = b + a := by
  exact Nat.add_comm a b
```

PageRank for Graph Analysis

Divide Mathematics fields

- Topology, Abstract Algebra, Linear Algebra, Calculus .etc
- heuristic categories based on namespace

Personalized PageRank (PPR)

- Calculate Personalized PageRank for each field

Subgraph PageRank

- Create subgraph of each fields, calculate PageRank

Full Top1000 PageRank

inductive: 294 / 1,000 = 29.400%
constructor: 269 / 1,000 = 26.900%
abbrev: 215 / 1,000 = 21.500%
definition: 177 / 1,000 = 17.700%
theorem: 40 / 1,000 = 4.000%
axiom: 3 / 1,000 = 0.300%
quotient: 2 / 1,000 = 0.200%

Explicit Top1000 PageRank

theorem: 335 / 1,000 = 33.500%
definition: 267 / 1,000 = 26.700%
abbrev: 217 / 1,000 = 21.700%
constructor: 161 / 1,000 = 16.100%
inductive: 14 / 1,000 = 1.400%
axiom: 3 / 1,000 = 0.300%
quotient: 3 / 1,000 = 0.300%

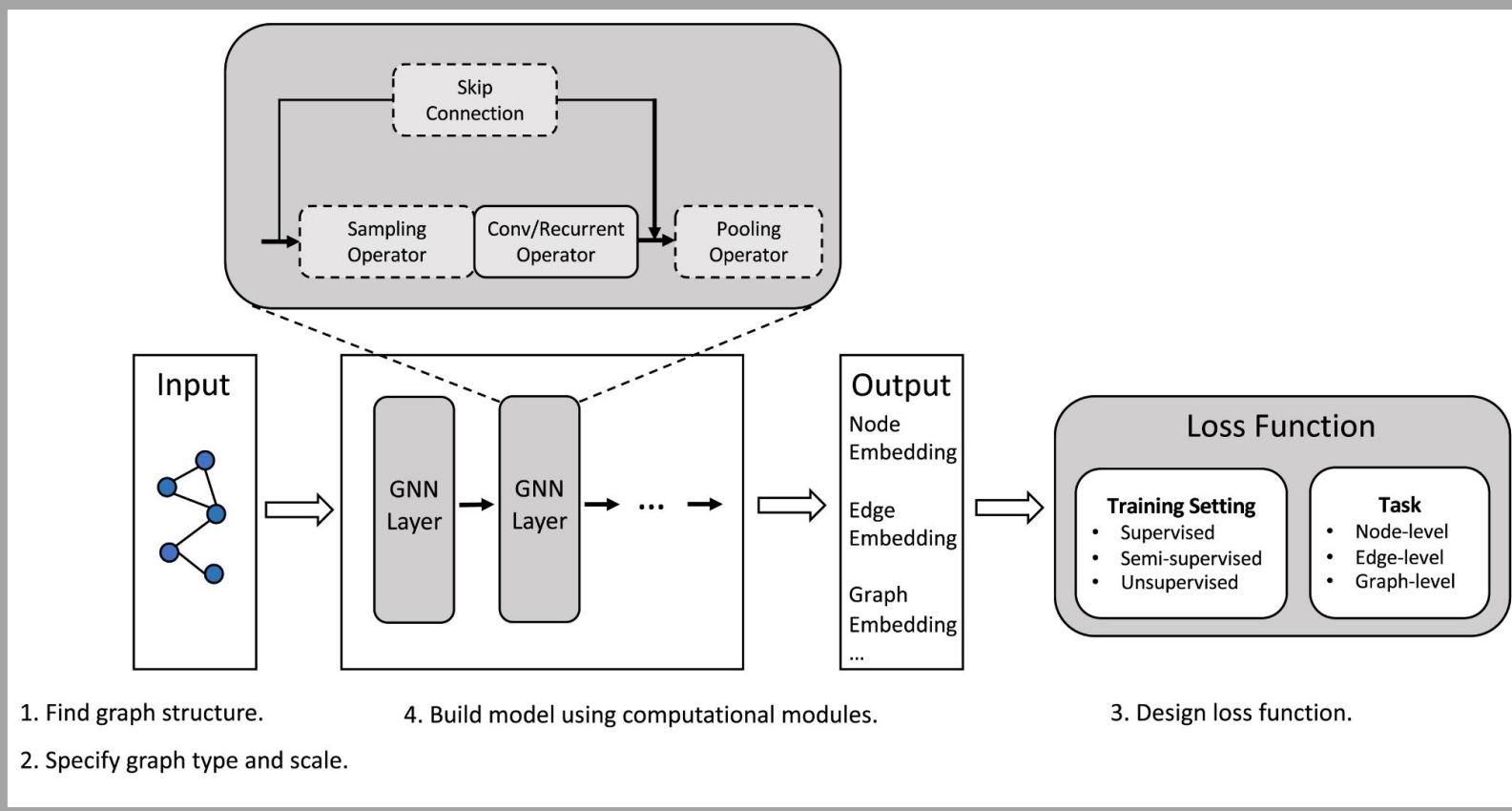
GNN for Link Prediction

Feature Embedding

- Numeric: Graph info. (degree, PageRank, etc.)
- Categorical: Kind (Thm, Def, Abbrev, etc.) one-hot vector.
- Semantics: Name vs. Type (Equation)

GNN Architecture

- GraphSAGE
- Dot Product + Concatenation
- Model Training



PageRank Results

PPR Limitation

- Almost similar for every fields! (DAG - Directed Acyclic Graph)

	Topology_Core	Abstract_Algebra	Linear_Algebra_Modules	Analysis_Normed_Calculus	Measure_Probability	Order_Lattice	Finite_Combinatorial_Graph	Number_Polynomial	CategoryTheory_Stress
Topology_Core	1.00	0.99	0.99	0.99	0.99	0.99	0.99	0.99	0.99
Abstract_Algebra	0.99	1.00	0.99	0.99	0.99	0.98	0.99	0.99	0.99
Linear_Algebra_Modules	0.99	0.99	1.00	1.00	1.00	0.98	1.00	0.99	0.99
Analysis_Normed_Calculus	0.99	0.99	1.00	1.00	1.00	0.98	1.00	0.99	0.99
Measure_Probability	0.99	0.99	1.00	1.00	1.00	0.99	1.00	0.99	0.99
Order_Lattice	0.98	0.98	0.98	0.98	0.98	1.00	0.98	0.98	0.98
Finite_Combinatorial_Graph	0.99	0.99	1.00	1.00	1.00	0.98	1.00	0.99	0.99
Number_Polynomial	0.99	0.99	0.99	0.99	0.99	0.98	0.99	1.00	0.99
CategoryTheory_Stress	0.99	0.99	0.99	0.99	0.99	0.98	0.99	0.99	1.00

Relative PageRank

- Relative PageRank = log(PPR / PageRank)
- Tendency to rely on DAG leaf node

	category	top_k	10	50	100	500	1000	5000	10000
Abstract_Algebra	0.000	0.000	0.000	0.000	0.000	0.120	0.497		
Analysis_Normed_Calculus	0.000	0.000	0.000	0.000	0.000	0.377	0.463		
CategoryTheory_Stress	0.000	0.000	0.000	0.000	0.000	0.000	0.000		
Finite_Combinatorial_Graph	0.000	0.000	0.000	0.000	0.000	0.000	0.417		
Linear_Algebra_Modules	0.000	0.000	0.000	0.000	0.000	0.031	0.434		
Measure_Probability	0.000	0.000	0.000	0.000	0.000	0.237	0.601		
Number_Polynomial	0.000	0.000	0.000	0.000	0.000	0.000	0.318		
Order_Lattice	0.000	0.000	0.000	0.000	0.173	0.171	0.104		
Topology_Core	0.000	0.000	0.000	0.000	0.000	0.293	0.410		

GNN Results

Test metrics

- ROC-AUC, MRR, Hit@N

Test ROC-AUC Score: 0.9957

Test for 843636 edges

MRR : 0.0055
Hit@5 : 0.0069
Hit@10 : 0.0109

Initial model

- Low score

Test for 843636 edges

MRR : 0.0055
Hit@5 : 0.0069
Hit@10 : 0.0109

Find Proper model

- Minimum validation loss model vs. More trained model

Evaluation portion: 10000/217853

MRR : 0.0039
Hit@5 : 0.1151
Hit@10 : 0.2110
Hit@50 : 0.4690
Hit@100 : 0.6017

Evaluation portion: 10000/217853

MRR : 0.0831
Hit@5 : 0.1165
Hit@10 : 0.2086
Hit@50 : 0.4530
Hit@100 : 0.5821

Premise Recommendation & Conclusion

Observation

- Explicit graph > Full graph in Hit-rate
- Not just name-similar ones, **semantically close**

Limitation

- Similar declarations are not always premises.

Target declaration	True premises	Hits in Top-50	Hit ratio	Recall
Matrix.det_mul	46	18	36.00%	39.10%
Polynomial.degree_mul	18	12	24.00%	66.70%
deriv_sin	3	3	6.00%	100.00%
Nat.gcd_comm	4	4	8.00%	100.00%
Continuous.comp	3	2	4.00%	66.70%

Rank	Recommended premise	premise ?
9	HasDerivAt.sin	Yes
10	HasDerivAt.cos	No
14	HasDerivAt.cosh	No
15	HasDerivAt.sinh	No
18	DifferentiableAt.hasDerivAt	Yes
21	HasDerivAt.deriv	Yes
22	HasDerivWithinAt.sin	No
24	HasDerivAt.csin	No
39	HasDerivAt.ccos	No
40	HasDerivAt.comp	No

Further Steps, References

Further Steps

- Add Mathematical field-aware graph features (like subgraph PageRank)
- Use hard negative sampling
- extend declaration-level to tactic-state-level (like LeanDojo)

References

- [1] X. Li et al., "The Network Structure of Mathlib", arXiv preprint
- [2] K. Yang et al., "LeanDojo: Theorem Proving with Retrieval-Augmented Language Models", Advances in Neural Information Processing Systems (NeurIPS)
- [3] T. Zhu et al., "Premise Selection for a Lean Hammer", arXiv preprint
- [4] Z. Wang et al., "Tree-based premise selection for Lean4", Advances in Neural Information Processing Systems (NeurIPS), 2025.
- [5] G. Gao et al., "A semantic search engine for Mathlib4", Findings of the Association for Computational Linguistics (EMNLP), 2024.